

Scalable Multi-Agent Reinforcement Learning for Adaptive Traffic Signal Control

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Abstract—Multi-agent reinforcement learning (MARL) is increasingly used for adaptive traffic signal control, but reported gains can depend strongly on baseline quality, demand regime, network scale, and evaluation design. This paper presents a reproducible evaluation of a deliberately simple independent-agent architecture in which each signalised intersection is controlled by either a Deep Q-Network (DQN) or Proximal Policy Optimisation (PPO) agent in SUMO. The final study contains 175 deterministic evaluation runs and compares five controllers: an original fixed-time plan, a Webster-derived fixed-time plan, actuated control, independent DQN, and independent PPO. Demand sensitivity is evaluated on a 2×2 network under low, medium, severe high, and dynamic demand, while scalability is assessed under calibrated medium demand on 2×2 , 3×3 , 4×4 , and 5×5 networks containing 4–25 controlled intersections. Paired tests use matched seeds, Holm–Bonferroni correction, confidence intervals, and paired effect sizes. The results show that baseline strength materially changes the interpretation of MARL performance. DQN is statistically competitive with actuated control across the five primary metrics on the 2×2 medium-demand network, but from 3×3 onward it develops significant delay and queue disadvantages while preserving a similar completion rate. PPO scales less favourably and deteriorates substantially under severe oversaturation. Inference remains computationally lightweight over the tested range: at 25 agents, mean joint decision latency is 1.04 ms for DQN and 1.57 ms for PPO. The findings support independent DQN as a viable research architecture within the tested scale, but do not support universal MARL superiority over strong conventional adaptive control.

Index Terms—multi-agent reinforcement learning, adaptive traffic signal control, DQN, PPO, SUMO, scalability, reproducibility, intelligent transportation systems

I. INTRODUCTION

Urban traffic signal control is a networked sequential decision problem: a phase decision changes immediate discharge, residual queues, downstream arrivals, and the state seen at the next control step. Reinforcement learning (RL) is therefore a natural candidate for adaptive control, while multi-agent reinforcement learning (MARL) distributes the control problem across multiple intersections. Reviews nevertheless show substantial heterogeneity in network design, state and reward formulation, baselines, simulators, and evaluation procedures [1], [2].

A recurring evaluation problem is baseline strength. Large improvements over an arbitrary cyclic plan do not establish superiority over demand-informed fixed timing or a competent adaptive controller. Cabrejas-Egea *et al.* explicitly demonstrated the importance of comparison with stronger operational systems

[3]. Scalability is similarly multidimensional: an implementation may instantiate more agents while traffic efficiency or computational requirements deteriorate. Large-scale MARL studies use communication, parameter sharing, graph methods, or distributed architectures to address these challenges [4]–[7].

This work asks a narrower question: how far can a simple independent-agent DQN/PPO architecture be generalised before behavioural or computational limitations become visible? The learned controllers follow the DQN formulation of Mnih *et al.* [8] and the PPO formulation of Schulman *et al.* [9]. The contribution is not a new RL algorithm. Instead, it is an evaluation framework that combines (i) stronger conventional baselines, (ii) multiple demand regimes, (iii) scaling from 4 to 25 controlled intersections, (iv) matched-seed statistical analysis, and (v) explicit computational measurements. This makes it possible to distinguish architectural scalability from traffic-performance scalability and inference feasibility.

II. RELATED WORK

Traffic-specific RL design has evolved beyond generic state-action formulations. IntelliLight demonstrated deep RL using traffic-domain state and reward design [10]; PressLight linked learning to max-pressure principles [11]; and FRAP represented competition between traffic phases explicitly [12]. These approaches show that state, action, and reward definitions embed important domain assumptions.

Coordination is central when multiple intersections act simultaneously. Early work used coordination graphs and message passing [13]; CoLight introduced graph-attention communication [4]; and decentralised A2C was evaluated on large synthetic and Monaco networks [5]. At substantially larger scale, “Toward a Thousand Lights” studied decentralised RL across networks containing thousands of signals [6], while GPLight dynamically groups agents and shares parameters within groups [7]. These methods motivate the present comparison with a simpler architecture that avoids a shared critic, parameter sharing, and explicit inter-agent messaging.

Scalability also includes simulation and deployment constraints. CityFlow was designed for efficient multi-agent city-scale traffic simulation [14]. TinyLight treats storage and computation as deployment constraints [15], while SafeLight introduces explicit safety objectives [16]. The present work remains simulation-based and does not claim field-deployment

readiness; instead, it reports the computational cost that was directly measured for the implemented system.

III. METHODOLOGY

A. Simulation and Independent-Agent Architecture

Experiments use the Simulation of Urban MObility (SUMO) microscopic simulator with Python control through TraCI [17]. The learned-controller implementation uses Stable-Baselines3 [18]. Each signalised intersection is represented by one independent RL agent. DQN agents own separate networks and replay buffers; PPO agents own separate policy/value parameters and rollout buffers. There is no parameter sharing or centralised critic.

Each agent receives a six-element local observation: halted-vehicle count, accumulated lane waiting time, mean lane occupancy, encoded signal phase, elapsed green time, and mean queue condition of directly connected traffic-light neighbours. The action space is binary: *HOLD* the current green or request a *SWITCH* after the minimum-green constraint. The final controller enforces a 10-s minimum green, a 3-s yellow transition, and a 5-s decision interval.

The local reward combines waiting-time change, queue size, local discharge, and switching cost,

$$r_{i,t} = -\alpha \Delta W_{i,t} - \beta Q_{i,t} + \delta T_{i,t} - \lambda C_{i,t}, \quad (1)$$

with $\alpha = 0.3$, $\beta = 0.2$, $\delta = 0.1$, and $\lambda = 0.5$. Equation (1) makes the optimisation trade-off explicit: persistent delay and queues are penalised, discharge is rewarded, and unnecessary phase changes incur a switching cost. Fig. 1 summarises how these local observations, rewards, and HOLD/SWITCH actions connect the independent agents to SUMO through TraCI; the key architectural point is that the per-agent interface remains fixed as the number of controlled intersections grows.

B. Controllers and Demand Scenarios

Five controllers are evaluated: Original Fixed-Time, Webster-derived Fixed-Time, Actuated, DQN, and PPO. The stronger fixed-time baseline is generated using Webster-style demand-informed timing [19], and the actuated controller provides a non-learning adaptive reference consistent with conventional detector-responsive control [20].

The primary 2×2 network is tested under four planned-demand regimes: 600 vehicles (low), 1,500 (medium), 3,000 (severe high-demand stress test), and 1,200 in a temporally changing dynamic route. Scalability is tested under independently calibrated medium-demand route sets: 1,500 vehicles on 2×2 , 1,800 on 3×3 , 2,000 on 4×4 , and 1,566 on 5×5 . Calibration is based on comparable fixed-time service severity rather than proportional vehicle count. Under the Webster reference, mean waiting time remains between 15.02 and 16.10 s and completion remains between 92.36 and 94.27% across the four grid sizes.

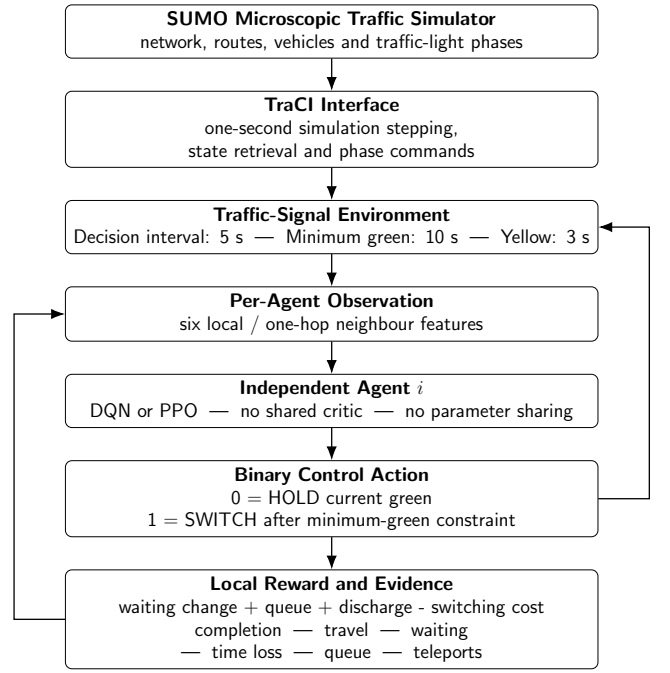


Fig. 1. Independent MARL system architecture used in the final experiment.

C. Training, Evaluation, and Statistics

Each learned controller is trained for 50,000 decision steps with training seed 1. Deterministic evaluation uses matched SUMO seeds 11–15. The final evidence base contains 100 unique demand-study runs plus 75 additional larger-grid medium-demand runs, for 175 unique evaluations.

Primary traffic metrics are completion rate, average travel time, average waiting time, average time loss, and mean sampled queue length. Seven predefined paired comparisons are performed within each scenario–metric or grid–metric family. Family-wise multiplicity is controlled using the Holm–Bonferroni procedure [21]. The analysis also reports paired mean differences, 95% confidence intervals, paired-samples Cohen’s d_z , and an exact paired sign-flip sensitivity analysis. For paired differences d_i , $d_z = \bar{d}/s_d$ [22], [23].

IV. RESULTS

A. Demand Sensitivity

Table I reports the main five-seed means. Under low and medium demand, actuated control provides the strongest descriptive traffic-efficiency results. At medium demand, DQN is close to actuated control: completion is 94.13% versus 94.43%, travel time is 98.48 s versus 96.62 s, and waiting time is 15.49 s versus 14.85 s. None of the five DQN–Actuated comparisons is Holm-significant at medium demand. PPO is significantly worse than Actuated for travel time, waiting time, time loss, and queue length.

The high-demand stress test changes the ranking substantially. Completion collapses for every controller; Webster is strongest descriptively at 15.53%. DQN remains comparatively robust

TABLE I
FIVE-SEED MEAN PERFORMANCE ACROSS THE FOUR 2×2 DEMAND REGIMES.

Scenario	Controller	Completion (%)	Travel (s)	Waiting (s)	Time loss (s)	Queue
Low	Original Fixed	94.93	97.43	19.57	33.80	1.627
Low	Webster	95.03	83.73	6.32	20.05	0.468
Low	Actuated	95.37	81.59	4.81	17.88	0.356
Low	DQN	95.17	83.15	6.11	19.43	0.383
Low	PPO	95.07	83.07	5.97	19.34	0.377
Medium	Original Fixed	92.95	122.23	36.24	59.56	6.513
Medium	Webster	94.27	98.89	15.13	36.28	2.315
Medium	Actuated	94.43	96.62	14.85	33.95	2.128
Medium	DQN	94.13	98.48	15.49	35.85	2.192
Medium	PPO	94.12	100.96	17.07	38.34	2.511
High	Original Fixed	13.44	273.54	162.00	200.89	74.092
High	Webster	15.53	258.63	134.48	187.96	70.123
High	Actuated	13.17	288.17	177.36	215.33	73.682
High	DQN	14.55	261.99	139.49	189.10	72.385
High	PPO	7.45	358.44	233.98	259.01	74.152
Dynamic	Original Fixed	93.02	116.10	28.95	50.18	4.334
Dynamic	Webster	93.95	102.84	14.54	36.87	2.005
Dynamic	Actuated	94.40	94.47	10.69	28.49	1.361
Dynamic	DQN	93.97	97.19	12.36	31.22	1.597
Dynamic	PPO	93.28	104.20	16.09	38.28	2.149

TABLE II
MEDIUM-DEMAND SCALABILITY (FIVE-SEED MEANS).

Agents	Controller	Compl. (%)	Travel	Waiting	Queue
4	Actuated	94.43	96.62	14.85	2.128
4	DQN	94.13	98.48	15.49	2.192
4	PPO	94.12	100.96	17.07	2.511
9	Actuated	93.73	107.20	14.28	1.324
9	DQN	93.49	110.99	16.48	1.444
9	PPO	93.28	115.63	19.68	1.753
16	Actuated	93.12	116.34	12.85	0.767
16	DQN	92.95	120.53	15.73	0.862
16	PPO	92.47	124.13	18.28	1.033
25	Actuated	92.81	125.43	12.07	0.385
25	DQN	92.57	129.50	15.20	0.417
25	PPO	92.48	131.49	16.64	0.469

at 14.55%, whereas PPO falls to 7.45% with 358.44 s average travel time and 233.98 s waiting time. DQN significantly outperforms PPO on completion, travel time, and waiting time under this severe regime. Under dynamic demand, DQN significantly outperforms Original Fixed-Time and Webster on several delay metrics but remains behind Actuated.

B. Behavioural Scalability

Table II summarises medium-demand scaling. Actuated control has the strongest descriptive completion, travel, waiting, and queue results at every tested grid size. DQN completion remains statistically indistinguishable from Actuated at 9, 16, and 25 agents, but significant delay disadvantages emerge from 9 agents onward. At 25 agents, DQN is 4.06 s slower than Actuated ($p_H = 0.0054$), has 3.12 s more waiting time ($p_H = 0.0018$), and a small queue difference of +0.033 ($p_H = 0.0478$). PPO generally scales less favourably.

C. Computational Scalability

Table III reports the directly measured computational quantities for all eight learned-controller/scale combinations. Training time increases with agent count for both algorithms, while

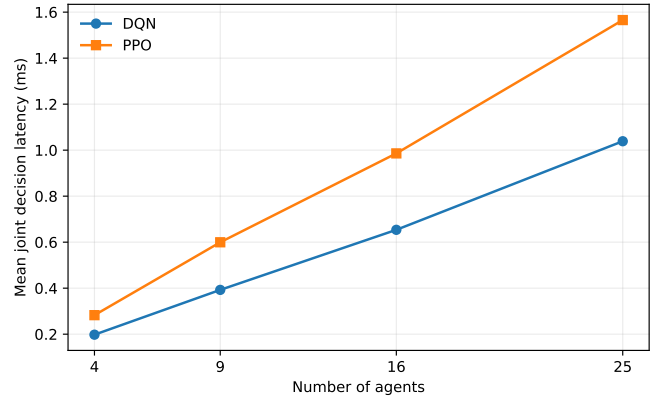


Fig. 2. Mean deterministic joint decision latency across the tested agent counts.

the saved model footprint remains small: at 25 agents it is 2.304 MB for DQN and 3.320 MB for PPO. DQN also shows a larger increase in measured RAM, reaching 422.66 MB at 25 agents, whereas PPO remains near 343 MB over the tested range.

Inference remains small compared with the 5-s decision interval. Fig. 2 shows that mean joint deterministic decision latency grows smoothly with the number of independent agents, but remains far below the control budget throughout the measured range. At 25 agents, the mean latency is 1.039 ms for DQN and 1.566 ms for PPO, corresponding to only 0.0208% and 0.0313% of the 5,000-ms decision interval. CPU/GPU utilisation was not instrumented during the frozen experiment and is therefore not reconstructed retrospectively; the computational conclusions are limited to the runtime, RAM, storage, and latency quantities reported in Table III.

TABLE III
MEASURED COMPUTATIONAL SCALABILITY OF INDEPENDENT DQN AND PPO.

Method	Agents	Training time (s)	Simulation runtime (s)	RAM (MB)	Model size (MB)	Mean latency (ms)
DQN	4	357.24	2.02	343.30	0.369	0.1978
DQN	9	692.93	3.96	357.94	0.829	0.3926
DQN	16	1046.14	6.84	388.56	1.475	0.6537
DQN	25	1398.06	10.63	422.66	2.304	1.0388
PPO	4	388.05	2.01	335.06	0.531	0.2827
PPO	9	731.54	4.05	334.73	1.195	0.5995
PPO	16	1018.70	6.95	339.75	2.125	0.9859
PPO	25	1469.70	10.79	343.27	3.320	1.5655

V. DISCUSSION

The main result is methodological as much as algorithmic: the apparent benefit of MARL changes with comparator strength. DQN and PPO produce large gains relative to the weak Original Fixed-Time plan in several conditions, yet Actuated control is often the strongest conventional comparator. This supports calls for stronger baselines in RL traffic-signal research [3] and illustrates why percentage improvement over a single static controller can be misleading.

DQN is consistently stronger than PPO in the more difficult parts of the experiment. Its medium-demand results are statistically competitive with Actuated on the smallest grid, it preserves completion rate comparatively well as the network grows, and it remains substantially more robust than PPO in the severe high-demand stress test. However, the growing delay gap against Actuated from nine agents onward indicates a coordination limitation. Independent agents operate with local and one-hop information and cannot optimise a global network objective directly. More coordinated methods such as graph-attention MARL, value decomposition, grouped parameter sharing, or centralised-training/decentralised-execution architectures are therefore natural comparisons for future work [4], [7], [24], [25].

The scaling result must also be bounded carefully. The implementation generalises from 2×2 to 5×5 networks while retaining a fixed local observation and action interface, but the experiment does not establish unbounded or city-scale scalability. Larger grids are evaluated only under calibrated medium demand; the severe and dynamic regimes are limited to 2×2 . Moreover, the networks are regular synthetic grids rather than heterogeneous real-city topologies. These limitations are important in light of work that explicitly models network heterogeneity [24] and much larger signal systems [6], [7].

VI. CONCLUSION

This paper evaluated a simple independent-agent MARL traffic-signal controller across multiple demand regimes and a 4–25-agent scalability range using strong conventional baselines and reproducible statistical comparisons. The results do not support universal MARL superiority. Actuated control is the strongest descriptive medium-demand controller at every tested network size. Independent DQN is nevertheless competitive with Actuated on the 2×2 medium network, preserves completion rate as the network grows, and is substantially more

robust than PPO under severe oversaturation. Deterministic inference remains far below the available decision-time budget throughout the tested range.

The principal implication is that MARL traffic-signal research should separate three questions: whether an architecture can instantiate more agents, whether traffic performance remains competitive as the network grows, and whether the computational cost remains practical. Strong baselines, multiple demand regimes, matched-seed statistics, and explicit resource measurements are necessary for credible claims. Future work should test additional training seeds, larger and irregular networks, multiple demand regimes at scale, explicit cooperative MARL, reward ablation, noisy observations, and controller-in-the-loop validation before field deployment.

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